ANURAG PANDEY

AI/ML & 3D Computer Vision Engineer | Backend Systems
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SKILLS

Languages: Python, Go, C++, CUDA, JavaScript/TypeScript, Dart, SQL
AI & 3D Vision: PyTorch, Open3D, OpenCV, MuJoCo, FoundationPose, SAM 2/3, DINOv2, ViT
Backend & Cloud: FastAPI, Flask, Django/DRF, Go WebSockets, React, Next.js, Three.js, Flutter, WebRTC, PostgreSQL, Redis, AWS (EC2, S3), Docker, Nginx, Linux

WORK EXPERIENCE

Terafac Technologies Pvt. Ltd. *Chandigarh, India*
Jr. Software Engineer – AI/ML & Vision *Jul 2026 – Sep 2026*
AI/ML & Vision Intern *Jan 2026 – Jul 2026*
AI/ML & Vision Engineer (Contract) *Jun 2024 – Dec 2025*

- Led development of core 3D vision (`vision-301`), cloud/edge backend (`api-103`), and 6D pose estimation (`vision-106`) deployed across 5 industrial robotic welding sites.
- Built a closed-loop **Vision-Tactile 6D Pose Calibration** pipeline combining RealSense RGB-D, FoundationPose, Kabsch/SVD plane fitting, and sub-pixel STag fiducial PnP drift compensation, **cutting seam error by 85–90% (4.5mm → 0.3mm)** in 18–22s.
- Engineered a coarse-touch alignment and analytical IK follower tightening probe search windows to 12 mm, **cutting calibration cycle time by 79% (282s → 58s)**.
- Architected `api-103` with batched 64-ray J6 hemisphere raycasting, optimized FoundationPose & SAM 3 microservices for **4 GB VRAM / Jetson Orin NX** (**25**× mask compression), and built the `obj_maker` Gemini CAD/STEP compiler.

AttentionVerse (TheCoder) *Delhi, India*
Software Engineer (Contract) – Backend & AI Systems *Jan 2022 – Apr 2024*

- Built `thecoder-backend` and `thecoder-ai` (Django, Celery, Redis, OpenAI) to dynamically generate live REST APIs, database schemas, and Nginx reverse proxies from natural language prompts.
- Developed `thecoder-search-2`, an AST-based semantic GitHub code search engine using Python `ast`, LangChain, and **ChromaDB**.
- Built the **IIT Mandi iHUB Health API** (Django REST, PostgreSQL, OpenCV OCR, Firebase Push) to scan medical reports and alert doctors on abnormal vitals.

Samsung Display Noida Pvt. Ltd. *Noida, India*
Assistant Engineer (Manufacturing / OLB Core) *Apr 2021 – Jan 2022*

- Worked with South Korean engineers in the Outer Lead Bonding (OLB) unit on machine troubleshooting, PLC diagnostics, and display panel yield optimization.

PROJECTS & COMPETITIONS

DexGraspVLA MuJoCo Simulation Bridge | *MuJoCo, PyTorch, SAM 2, Cutie, DINOv2* *Nov 2025 – Dec 2025*

- Ported the DexGraspVLA policy to MuJoCo with a **7-DoF Panda arm + 18-DoF Shadow Hand**, custom **6D-to-18D kinematic mapping**, SAM 2 + Cutie mask tracking, and a live DINOv2 attention heatmap visualizer.

Chess Stream Arena & NPC World | *Go, Flutter, LiveKit WebRTC, FastAPI, pgvector, Godot* *GitHub | 2026*

- Built **Chess Stream** (Go matchmaking + Flutter) streaming 1v1 video and chess moves over **WebRTC Data Channels**; built **NPC World**, a 2D multiplayer engine where LLM NPCs persist episodic `pgvector` memories and 7-D social vectors.

SyncHub (Dew Storage) & Quizzer (Go Engine) | *Flask, Go, WebSockets, AWS S3* *GitHub | 2024 – 2025*

- Built **SyncHub**, an offline-first LAN object store auto-syncing local buckets to AWS S3, and **Quizzer-Backend**, a concurrent Go 1.23 WebSocket server using channels and mutex-guarded session hubs for multiplayer quizzes.

ML Competitions (FOOD-101, I2V, Amazon ML) | *PyTorch, ViT, DeBERTa-v3, SE-ResNet-50, LightGBM* *2025 – 2026*

- **FOOD-101:** Reached **84.24% Top-1 accuracy** via a ViT + SE-ResNet-50 ensemble. **I2V Challenge:** Achieved **86.80% F1** on 12-class vehicle detection with ONNX export. **Amazon ML:** Built a DeBERTa-v3 + ViT + LightGBM pricing model with custom `SMAPELoss`.

EDUCATION & LEADERSHIP

University School of Automation & Robotics (USAR), GGSIPU *Delhi, India*
B.Tech in Artificial Intelligence & Machine Learning (CGPA: 7.43/10 | 8th-Sem Project: 36/40, 1st in Batch) *Aug 2023 – Aug 2026*
Cloud Computing Team Lead, AWS Cloud Club GGSIPU *May 2024 – Jan 2025*

Ambedkar Institute of Technology (BTE Delhi) *Delhi, India*
Diploma in Digital Electronics (Percentage: 80.98%) *2017 – 2020*